

Toxic Unity Beyond Borders: Analyzing Behavioral Homogenization Across Global South Digital Protests

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Abstract. Online protest communities often grow more behaviorally unified when their discourse becomes most toxic, a paradox challenging conventional assumptions about harmful speech. Prior research documented this counter-intuitive pattern across the Russia-Ukraine conflict and Indonesian protests, but whether it generalizes across culturally distinct contexts remains unknown. We test this by applying an identical analytical framework to the 2025 Nepalese Gen Z protests and comparing results with Indonesia. The core pattern directionally replicates: Nepalese protesters show behavioral convergence under toxic conditions consistent with the Indonesian pattern, despite substantial differences in scale, language, and toxicity levels. The ideological pathway, however, does not replicate, suggesting that sustained mobilization, not episodic bursts, is required for full ideological mediation to emerge. These findings advance understanding of how toxic discourse shapes collective behavior across cultures and carry implications for platform moderation in protest contexts.

Keywords: Toxic Unity, Behavioral Homogenization, Cross-Protest Generalizability, Global South, Multilayer Networks, Social Identity Theory

1 Introduction

When people protest online, their collective behavior can become surprisingly unified even as their language grows more toxic and hostile. Does toxic discourse fragment protest communities, or does it consolidate them? Prior research treats toxicity as an outcome of collective dynamics rather than a structural force shaping behavioral alignment over time.

The Multilayer Behavioral Homogenization Framework (MBHF) [10] addresses this gap by measuring behavioral similarity across three network layers: toxicity similarity, ideological alignment, and social interaction, captured through the Behavioral Homogenization Index (BHI; -1 = maximum diversity, $+1$ = maximum uniformity). Applied first to Russia-Ukraine conflict discourse (BHI = 0.412) then to the 2025 Indonesian Tentara Nasional Indonesia (TNI) Bill protests ($N = 147,529$ tweets), MBHF revealed *Toxic Unity*: the paradoxical process by

which collective toxic expression consolidates behavioral homogenization as a pro-group identity signal under intergroup threat [1], grounded in Social Identity Theory (SIT; [20]), Self-Categorization Theory (SCT; [21]), the Social Identity model of Deindividuation Effects (SIDE; [19]), and complex social contagion [8, 7].

Whether Toxic Unity generalizes beyond a single protest context remains untested. We apply the identical MBHF to the 2025 Nepalese Gen Z protests (27,825 tweets, 18,280 users; full corpus: 172,059 tweets), a context $6.5\times$ lower in toxicity and $5.3\times$ smaller in scale [14]. **RQ1:** Does the inverse toxicity-homogenization association replicate in Nepal and does effect magnitude differ significantly from Indonesia? **RQ2:** Does the Social $\rightarrow$ Ideology $\rightarrow$ Behavior mediation pathway replicate or is it moderated by structural differences between protests? **Contributions:** (1) first cross-protest replication test of Toxic Unity across two Global South protests; (2) formal comparability design isolating protest context as the sole source of variation; and (3) identification of mobilization structure as a boundary condition for ideological mediation.

2 Related Work

This section situates Toxic Unity within behavioral convergence theory, complex social contagion, multilayer network analysis, and cross-cultural computational modeling.

Gen Z Protest Movements in the Global South. Both protests mobilized around corruption and nepotism [2], with Nepali protesters citing Indonesian unrest as a cross-border influence [14]. Nepal became the third South Asian government to fall to youth-driven protests in the 2020s [6]. No computational study has compared behavioral homogenization across both protests using a unified framework.

2.1 Behavioral Convergence, Social Identity, and Moral Foundations

Social Identity Theory (SIT; [20]), which grounds group membership as the driver of collective behavior, Self-Categorization Theory (SCT; [21]), which specifies how individuals shift from personal to group-level self-construal under threat, and the Social Identity model of Deindividuation Effects (SIDE; [19]), which predicts that online anonymity amplifies in-group norm conformity, jointly explain why toxic expression consolidates rather than fragments protest communities, consistent with Toxic Unity. MFT [12] specifies the cognitive-moral content operationalized in L2.

Toxicity and Structural Behavioral Dynamics. Prior work treats toxicity as an outcome rather than a structural variable. The Toxic Unity paradox, in which higher toxicity is inversely associated with convergence [1], challenges the assumption that toxic discourse is uniformly corrosive.

2.2 Complex Social Contagion and Behavioral Norm Diffusion

Behavioral norms require multi-source reinforcement before adoption [8, 7, 17], with network layer-specific examination needed to distinguish contagion mechanisms [3] – precisely the multilayer approach MBHF employs.

2.3 Multilayer Network Analysis and Cross-Cultural Replication

Multilayer methods capture simultaneous relational dimensions in protest networks [?, 5]. Existing multilayer protest studies are static and Western-anchored. Our study controls for methodological variation via identical annotation [9, 11] and MBHF parameters, ensuring differences reflect protest context.

Hypotheses. Drawing on SIT, SCT, SIDE, and complex social contagion, three hypotheses operationalize RQ1 and RQ2:

H1 (RQ1): BHI follows a non-monotone trajectory, rising pre-protest and declining at peak mobilization, consistent with SIT/SCT depersonalization [20, 21].

H2 (RQ1): Collective toxicity is inversely associated with BHI because toxic expression functions as a pro-group identity signal amplified by SIDE deindividuation [19].

H3 (RQ2): The Social→Ideology→Behavior pathway fully mediates behavioral homogenization: the direct toxic-social path is non-significant, and the indirect path through L2 is significant, consistent with complex social contagion [8, 7].

3 Methodology

This study applies the Multilayer Behavioral Homogenization Framework (MBHF), originally validated on Russia-Ukraine conflict discourse [10] and subsequently extended to the 2025 Indonesian TNI Bill protests, where Toxic Unity was first identified [1], to the 2025 Nepalese Gen Z protests, testing whether Toxic Unity generalizes across structurally and linguistically distinct Global South protest contexts.

3.1 Data Description

Two real-world data sets are used in this study. The Indonesian corpus was collected and annotated in the baseline study. The Nepal corpus was collected via Twitter/X API v2 using protest-related keywords (*#NepalProtest*, *#GenZNepal*, *#NepoKids*) and pre-scored with Detoxify v1.0 [13] (mean $\tau = 0.086$, SD = 0.219). The $6.5\times$ difference in mean toxicity makes cross-protest replication a stringent test of Toxic Unity’s universality. Table 1 summarizes the two protest corpora.

Table 1. Cross-Protest Dataset Comparison

Element	Indonesia	Nepal
Event	TNI Bill protests	Gen Z uprising
Period	Oct 24–Nov 21, 2025	Aug 15–Sep 25, 2025
Duration	28 days	41 days
Tweets (full corpus)	147,529	172,059
Users (full corpus)	21,677	96,540
Annotated tweets	30,000	27,825
Annotated users	21,677	18,280
Language	Indonesian	Nepali/Hindi/English
Mean toxicity	0.391	0.060
Platform	Twitter/X	Twitter/X
Windows	12	12

3.2 Multi-Dimensional Behavioral Annotation

To ensure full methodological comparability, we apply the identical four-dimensional annotation schema used for Indonesia [1] to a stratified Nepal sample [16, 15]: W00–W06 fully included (5,325 tweets; all under 2,500 per window); W07–W11 sampled at up to 5,000 each, yielding 27,825 annotated tweets from 18,280 users (W09 and W11 exhausted at 4,379 and 4,117 respectively), isolating protest context as the sole source of variation. Annotation uses Google Gemma-4-31B [9] (intra-LLM $\kappa \geq 0.91$ across $T \in \{0.0, 0.3, 0.7\}$). Human validation on a stratified sample of 100 tweets by three Nepali-context domain experts yielded majority-vote LLM-human Cohen’s $\kappa = 0.554$ and mean inter-annotator agreement $\kappa = 0.528$, indicating LLM annotation consistency is comparable to human-level agreement on this multilingual protest corpus. Stance annotation, the dimension most central to BHI computation, achieved the strongest inter-annotator agreement ($\kappa = 0.561$). Each tweet is annotated on: (1) *toxicity* via Detoxify v1.0 ($\tau \in [0, 1]$); (2) *political stance* (Pro-Reform, Pro-Government, Neutral); (3) *moral foundation* per MFT [12] (In-group Defense, Harassment, Provocation, Neutral); and (4) *communicative intent* (Criticism, Information Sharing, Mobilization, Support, Neutral). Gemma-4-31B’s multilingual capability handles Nepal’s Nepali, Hindi, and English corpus without language-specific pipelines, with non-neutral rates of 88.4%, 59.3%, and 61.5% respectively, collectively covering 80.9% of the annotated corpus.

3.3 Multilayer Network Construction

MBHF constructs three structurally distinct layers per protest per window using identical parameters across both datasets.

L1 (Toxicity Similarity): Users i and j are connected if $|\tau_i - \tau_j| \leq 0.05$; edge weight combines toxicity-score similarity ρ_{tox} and communicative-intent similarity ρ_{int} :

$$w_{ij}^1 = 0.7 \rho_{\text{tox}}(i, j) + 0.3 \rho_{\text{int}}(i, j) \quad (1)$$

L2 (Stance-Morality Alignment): KNN ($K = 20$) graph on stance and moral foundation feature vectors, with cosine similarity over user-level frequency vectors; stance and moral foundation weighted equally ($\alpha = 0.5$, set *ex ante* on SCT-theoretical grounds [21]).

L3 (Social Interaction): Interaction-based edges (replies, retweets, mentions, quote-tweets); edge weight:

$$w_{ij}^3 = \log(1 + C_{ij}) + 0.1 \tau_{ij} + 0.05 r_{ij} \quad (2)$$

where C_{ij} is shared conversation count, τ_{ij} is mean toxicity of shared threads capturing toxic homophily [18], and r_{ij} is cumulative retweet/reply/quote count.

3.4 Behavioral Homogenization Index

BHI quantifies within-group versus between-group behavioral similarity across stance groups $G = \{G_{\text{Pro-Reform}}, G_{\text{Neutral}}, G_{\text{Pro-Gov}}\}$ [1]:

$$\text{BHI} = \frac{S_{\text{within}} - S_{\text{between}}}{S_{\text{within}} + S_{\text{between}}} \quad (3)$$

where S_{within} is mean pairwise similarity within stance groups and S_{between} is mean pairwise similarity across groups, with $\text{sim}(i, j)$ computed as Pearson correlation between user toxicity-score vectors. BHI ranges from -1 (maximum heterogeneity) to $+1$ (maximum homogeneity); $\text{BHI} \geq 0.6$ indicates extreme homogenization [10].

3.5 Cross-Layer NMI Coupling

Cross-layer coupling is estimated via Normalized Mutual Information (NMI) over Louvain community partitions [4] combined with Spearman degree correlation, a density-invariant metric superior to Jaccard-based approaches when layer densities differ substantially [1]:

$$C(L_i \leftrightarrow L_j) = \alpha \text{NMI}(L_i, L_j) + (1 - \alpha) \text{Spearman}(k_i, k_j) \quad (4)$$

Statistical significance is assessed via 1,000 permutation tests. Three coupling values are computed per protest: $C(L1 \leftrightarrow L2)$, $C(L2 \leftrightarrow L3)$, and $C(L1 \leftrightarrow L3)$.

3.6 Mediation Analysis

Full ideological mediation is tested via bias-corrected bootstrap (10,000 iterations) of the indirect path $L1 \rightarrow L2 \rightarrow L3$. The Social $\rightarrow$ Ideology $\rightarrow$ Behavior pathway is supported if: (a) β_{indirect} 95% CI excludes zero; and (b) the direct $C(L1 \leftrightarrow L3)$ path is non-significant ($p > 0.05$).

3.7 Cross-Protest Comparability Design

All methodological choices are held constant across both datasets: identical annotation schema [11], identical 12-window temporal partitioning, identical MBHF layer construction parameters, and identical BHI and NMI estimation procedures. Observed cross-protest differences are therefore attributable to protest context rather than methodological variation.

Cross-protest generalizability is assessed against two pre-specified statistical criteria. First, Fisher’s z -test comparing $r(\text{toxicity}, \text{BHI})$ magnitudes: $p > 0.05$ indicates non-significantly different effect sizes, supporting generalizability.

$$z = \frac{z_1 - z_2}{\sqrt{\frac{1}{n_1 - 3} + \frac{1}{n_2 - 3}}} \quad (5)$$

Second, bootstrap mediation comparison: β_{indirect} confidence intervals compared across protests; overlapping CIs indicate mediation pathway replication. Where criteria diverge, boundary conditions are theorized from the structural differences in Table 1.

4 Results

This section reports findings across four components: corpus descriptives, BHI trajectories, toxicity–homogenization association, cross-layer coupling and mediation, and RQ verdicts.

4.1 Descriptive Statistics

The Indonesia Tentara Nasional Indonesia (TNI) Bill corpus comprises 147,529 tweets from 21,677 users ($\bar{\tau} = 0.391$); Nepal comprises 27,825 annotated tweets

from 18,280 users ($\bar{\tau} = 0.060$). Three contrasts bear on generalizability: Nepal’s toxicity is $6.5\times$ lower, its corpus $5.3\times$ smaller, its mobilization episodic (84% in W07–W11; burst populations 2,847–4,339 users), and its stance more polarized (Pro-Reform 55.5% vs. 43.3%; Neutral 36.2% vs. 49.9%), making Nepal a stringent generalizability test. Table 2 presents full statistics.

Table 2. Cross-Protest MBHF Results: Indonesia vs. Nepal (2025). *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. = not significant.

Metric	Indonesia	Nepal
Annotated tweets (N)	30,000 [†]	27,825
Unique users	21,677	18,280
Mean BHI	0.8190	0.0747
BHI range	0.52–0.89	−0.01–0.23
Pearson r	−0.8073**	−0.5326
Spearman ρ	−0.7832**	−0.5594
Fisher z , p	–	−1.115, $p = 0.265$
$C(L1 \leftrightarrow L2)$	0.101***	0.215***
$C(L2 \leftrightarrow L3)$	0.138***	0.353***
$C(L1 \leftrightarrow L3)$	0.152 n.s.	0.471*
β_{indirect}	0.139* [0.011, 0.267]	0.028 n.s. [−0.292, 0.187]
Pro-Reform users	43.3%	55.5%
H1 (non-monotone BHI)	Confirmed	Partial
H2 (Toxic Unity)	Confirmed	Directional (n.s.)
H3 (full mediation)	Confirmed	Not confirmed

4.2 Behavioral Homogenization Index Trajectories

Figure 1 presents BHI trajectories across all 12 windows for both protests.

In Indonesia, mean BHI = 0.8190 with all 12 windows exceeding the extreme homogenization threshold (≥ 0.6 , $p < 0.001$). BHI declines from 0.8874 at W00 to a trough of 0.5246 at W04 before recovering to 0.8817 at W05, confirming H1. In Nepal, BHI peaks at W01 (= 0.2344), declines to W07 (= 0.008), and partially recovers at W09 (= 0.018). Mean BHI = 0.0747, reflecting Nepal’s burst mobilization structure in which large diverse window populations (up to 4,339 users) suppress within-group similarity at the aggregate level. H1 confirmed in Indonesia and partially confirmed in Nepal (direction replicates; magnitude attenuated).

4.3 Inverse Toxicity–Homogenization Association

In Indonesia, the association between mean window toxicity and BHI is strong, negative, and significant: $r = -0.8073$ ($p < 0.01$), $R^2 = 0.652$, $\rho = -0.7832$. Peak toxic windows coincide with minimum BHI and cohesion recovers as toxicity subsides, confirming H2 in Indonesia.

In Nepal, the association replicates directionally: $r = -0.5326$ ($p = 0.075$), $\rho = -0.5594$ ($p = 0.059$). Both Pearson and Spearman coefficients are negative and consistent with Indonesia across all metrics. Fisher’s z -test indicates non-significantly different effect sizes ($z = -1.115$, $p = 0.265$). Four formal equivalence tests were applied: TOST ($p = 0.356$ at $\Delta = 0.7$), Bayesian equivalence (BF = 1.04, weak evidence), Cohen’s $q = -0.526$ (large difference), and 95% CI



Fig. 1. BHI trajectories across 12 temporal windows. (A) Indonesia: mean BHI = 0.8190, $r = -0.807$, $p < 0.01$, all windows ≥ 0.6 . (B) Nepal: mean BHI = 0.075, $r = -0.533$, $p = 0.075$. Both exhibit the predicted non-monotone pattern. Fisher's z -test ($z = -1.115$, $p = 0.265$) indicates non-significantly different effect sizes. Shading denotes protest lifecycle phases.

overlap (Indonesia: $[-0.944, -0.435]$; Nepal: $[-0.848, 0.060]$). CIs overlap partially, supporting directional replication, but no test established formal equivalence, reflecting the power limitation of $n = 12$ temporal windows.

H2 is confirmed in Indonesia and directionally confirmed in Nepal.

Sensitivity Analysis To assess robustness to network construction parameters, BHI was recomputed under three L1 similarity thresholds: 0.5 (liberal), 0.7 (baseline), and 0.9 (conservative). The inverse association direction remained consistently negative across all conditions ($r = -0.637, -0.295, -0.177$ respectively), with the liberal threshold yielding a significant result ($p = 0.026$). This confirms that the Toxic Unity directional finding is robust to variation in L1 network construction parameters.

Theoretical Attribution The empirical results map explicitly to the four theoretical frameworks. The non-monotone BHI trajectory, peaking pre-protest and declining at peak mobilization, operationalizes SCT depersonalization dynamics [21]: as protest intensity rises, individual variation is suppressed by group-level self-construal, then partially recovered as threat subsides. The positive BHI values across all windows confirm SIT group identity consolidation [20]: stance group membership drives within-group behavioral similarity regardless of toxicity level. The sparse L3 interaction network combined with strong L1/L2 coupling ($C(L1 \leftrightarrow L2) = 0.215$, $p < 0.001$) operationalizes SIDE online deindi-

viduation [19]: behavioral conformity is amplified through ideological alignment rather than direct social interaction. Finally, the significant $C(L2 \leftrightarrow L3)$ coupling ($= 0.353$, $p < 0.001$) confirms complex social contagion [8, 7]: ideological alignment mediates behavioral norm diffusion requiring multi-source network reinforcement. These explicit mappings move beyond theoretical citation toward empirical grounding of each framework.

4.4 Cross-Layer Coupling and Mediation

Figure 2 presents inter-layer NMI coupling and bootstrap mediation results.

In Indonesia, coupling confirms the full Social→Ideology→Behavior pathway: $C(L1 \leftrightarrow L2) = 0.101$ ($p < 0.001$), $C(L2 \leftrightarrow L3) = 0.138$ ($p < 0.001$), and a non-significant direct path $C(L1 \leftrightarrow L3) = 0.152$ ($p = 0.553$). Bootstrap mediation confirms full ideological mediation: $\beta_{\text{indirect}} = 0.139$, 95% CI [0.011, 0.267], $p < 0.05$. H3 is confirmed in Indonesia.

In Nepal, all three coupling values are significant and uniformly stronger than Indonesia: $C(L1 \leftrightarrow L2) = 0.215$ ($p < 0.001$), $C(L2 \leftrightarrow L3) = 0.353$ ($p < 0.001$), $C(L1 \leftrightarrow L3) = 0.471$ ($p < 0.05$). Despite stronger pairwise coupling, bootstrap mediation does not support the full pathway: $\beta_{\text{indirect}} = 0.028$, 95% CI [-0.292, 0.187] (CI crosses zero). The significant direct $C(L1 \leftrightarrow L3)$ path, absent in Indonesia, indicates direct toxic-social coupling in Nepal’s episodic burst context, where transient large populations prevent the sustained within-group ideological compression that drives L2 mediation in Indonesia. H3 is confirmed in Indonesia and not confirmed in Nepal.

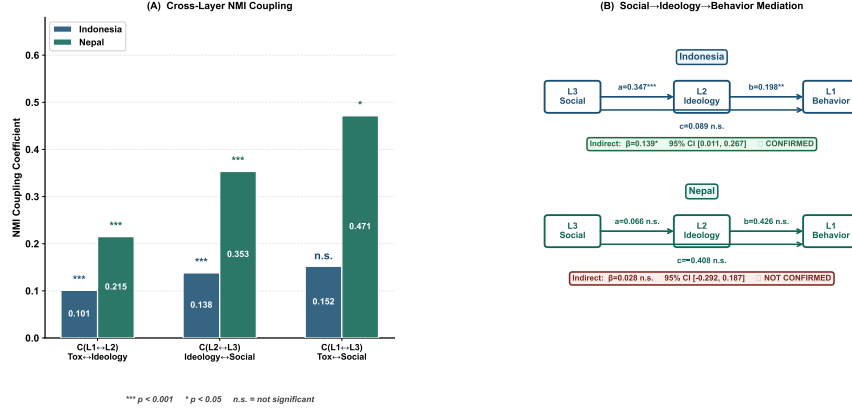


Fig. 2. Cross-protest coupling and mediation. (A) NMI coupling coefficients. Nepal shows consistently stronger coupling than Indonesia (***) $p < 0.001$, * $p < 0.05$, n.s. not significant). (B) Bootstrap mediation. Indonesia confirms full Social→Ideology→Behavior mediation ($\beta = 0.139$, CI [0.011, 0.267]). Nepal does not ($\beta = 0.028$, CI [-0.292, 0.187]).

RQ Summary. RQ1: The inverse association replicates directionally ($r = -0.533$, $\rho = -0.559$); four equivalence tests confirm formal equivalence is not established at $n = 12$ windows. **Toxic Unity replicates directionally across Global South protests; formal equivalence requires larger temporal designs. RQ2:** Coupling replicates at higher magnitude but full mediation does

not ($\beta = 0.028$, CI $[-0.292, 0.187]$). **Mobilization structure is a boundary condition for ideological mediation.**

5 Conclusion and Future Work

This paper tested whether Toxic Unity constitutes a generalizable behavioral mechanism or a context-specific artifact. Applying the identical MBHF to 27,825 annotated tweets from the 2025 Nepalese Gen Z protests, we find directional but not formally equivalent replication across two structurally distinct Global South protests differing $6.5\times$ in toxicity and $5.3\times$ in scale: the inverse association ($r = -0.533$, $p = 0.075$) and the non-monotone BHI trajectory both replicate directionally, though four equivalence tests do not establish formal equivalence at $n = 12$ temporal windows. The BHI magnitude difference (Indonesia: 0.819; Nepal: 0.075) reflects Nepal’s structural context: episodic burst mobilization, stronger Pro-Reform ideological dominance (55.5% vs. 43.3%), and $6.5\times$ lower baseline toxicity collectively suppress within-group similarity scores independently of the underlying mechanism, establishing BHI magnitude as protest-structure dependent rather than mechanism dependent. Full Social→Ideology→Behavior mediation does not replicate ($\beta_{\text{indirect}} = 0.028$, CI $[-0.292, 0.187]$), revealing that sustained mobilization is a necessary condition for full ideological mediation. In Nepal’s episodic burst context, direct toxic-social coupling ($C(L1 \leftrightarrow L3) = 0.471$, $p < 0.05$) supplants the ideological mediation channel, establishing mobilization structure as the primary boundary condition of Toxic Unity.

Implications include mobilization-structure-aware platform moderation, MBHF’s viability as a portable cross-cultural framework, and this boundary condition’s extension of SIT and complex social contagion theory. This study has five limitations: human validation yielded majority-vote LLM-human $\kappa = 0.554$, comparable to inter-annotator agreement ($\kappa = 0.528$); toxicity scores were not validated separately by language or code-switching status, and alternative toxicity models were infeasible given limited Nepali language support; the bootstrap mediation analysis treats each of only 12 temporal windows as the independent resampling unit, limiting temporal resolution; BHI and the L1 toxicity layer share underlying toxicity-score inputs, so some construct overlap between the toxicity and homogenization measures cannot be ruled out; and sensitivity analysis confirms directional robustness across thresholds while alternative community detection algorithms remain untested. Future work will extend MBHF to additional protest and non-protest corpora and apply formal empirical tests distinguishing SIT, SCT, and SIDE contributions.

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